

Modernizing Herschel’s Star Gauging: A Computational Approach Using Lick Observatory Data

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Abstract

This project revisits and modernizes William Herschel’s star gauging method using digital imaging and contemporary computational tools. Multiple automated star-counting approaches are developed, tested, and compared using images from Lick Observatory and related sources. The goal is to evaluate detection accuracy, identify methodological limitations, and assess how modern techniques refine star-count-based stellar density estimation.

1 Introduction

1.1 Context

William Herschel’s star gauging method tried to infer the large-scale structure of the Milky Way by counting stars in different locations in the sky. His approach relied on assumptions of uniform stellar distribution and constant detectability across the telescope’s field of view. While these assumptions are now known to be inaccurate, the underlying concept remains a useful framework for understanding galactic structure.

This project recreates and updates Herschel’s methodology using modern digital images, telescopes, and automated star detection algorithms. By comparing multiple computational approaches, this work evaluates the feasibility and accuracy of star-count-based density estimation with contemporary instruments.

1.2 Contributions

This project includes:

- Development of automated star-counting tools for Lick Observatory data
- Comparison of multiple computational detection methods
- Estimation of relative stellar density from detection counts
- Evaluation of accuracy, consistency, and methodological limitations

1.3 Goals

The report addresses the following questions:

- Which computational methods perform best for automated star counting?
- How consistent are these methods under realistic observational constraints?
- What challenges arise from sensor noise, resolution limits, and observing conditions?

2 Background

2.1 Star Gauging

Star gauging involves counting stars within defined fields of view to infer spatial variations in stellar density. Herschel’s manual counts led to one of the earliest conceptual models of the Milky Way. Despite its limitations, the method represents an important historical foundation of Galactic astronomy.

2.2 Modern Constraints and Data Challenges

Digital imaging introduces several challenges for automated detection:

- **Limiting magnitude:** faint sources fall below detection thresholds

- **Noise:** read noise, dark current, and sky background
- **Image resolution:** atmospheric seeing and undersampling
- **Sensor artifacts:** hot pixels, column defects, amplifier glow
- **Cosmic rays:** single-frame bright artifacts
- **Variability:** exposure differences and tracking errors

These all arise from the use of the types of telescopes (CCDs), and are difficult to fix, especially with issues on the Lick telescope along with other things on the Unistellars.

2.3 Computational Approaches

Common star detection techniques include:

- Threshold-based peak detection
- Aperture photometry and centroiding
- PSF-fitting and model-based extraction
- Segmentation algorithms (e.g., DAOSTarFinder)
- Machine learning approaches, including CNN-based classifiers

3 Data Sources and Preprocessing

3.1 Image Sources

The dataset consists of images from Lick Observatory and potentially Unistellar telescopes, spanning a range of fields, exposure times, and observing conditions. Calibration frames may accompany some datasets.

3.2 Raw Data Issues

Common issues in our data include:

- Noise and hot pixels
- Saturation and blooming
- Cosmic ray strikes
- Minor tracking errors and star elongation

3.3 Preprocessing

The following preprocessing steps were applied prior to detection:

- **Background Estimation:** Sigma-clipped statistics to estimate background mean, median, and noise
- **Background Subtraction:** Median background removal to improve contrast
- **Data Conversion:** FITS data converted to 32-bit floating point
- **Normalization (CNN only):** Pixel values scaled to $[0, 1]$
- **Dimensional Handling:** Automatic selection of the first 2D slice from multi-dimensional FITS files
- **Noise Estimation:** Both sigma-clipped standard deviation and MAD for robustness

Traditional calibration steps (bias, dark, flat correction, cosmic ray removal) were assumed to have been applied during earlier reduction stages. This project focuses on detection and classification rather than raw calibration.

4 Methods

4.1 Baseline Detection Methods

Thresholding with Connected Components: Percentile-based thresholding (95th percentile) combined with connected-component labeling was used to identify star-like regions. This approach:

- Applied binary masking and region labeling
- Filtered detections by minimum area
- Found centroids via center-of-mass calculations
- Visualized detections using contour overlays

Photutils Parameter Exploration: `DAOSTarFinder` was used to explore detection behavior across parameter ranges:

- Varied detection thresholds and FWHM values
- Used sigma-clipped background statistics
- Visualized detections with circular markers
- Logged results to text files

4.2 Photutils–GMM Pipeline

A more advanced pipeline combined traditional detection with machine learning:

- **Detection:** `DAOSTarFinder` with sigma-clipped background estimation
- **Feature Extraction:** Flux, peak intensity, second moments, and eccentricity
- **Classification:** Gaussian Mixture Model (GMM) applied to standardized features
- **Spatial Merging:** DBSCAN and single-link clustering to merge nearby detections

- **Output:** Diagnostic plots and CSV summaries with full detection metrics

DBSCAN was not that effective at fixing duplicate detections in crowded regions or in especially large objects, this will need to be improved later.

5 Conclusion

5.1 Overall Assessment

Among the tested methods, the Photutils–GMM pipeline provided the best combination of detection accuracy, robustness, and efficiency. By combining existing tools with machine learning classification, this method improved upon simpler techniques.

Key strengths include:

- Reliable detection in moderately crowded fields
- Improved discrimination between stars and noise artifacts
- Efficient performance relative to connected-component methods
- Detailed diagnostic outputs for further analysis

5.2 Relation to Herschel’s Method

The progression from manual counting to automated, feature-based pipelines reflects the methodological evolution since Herschel’s era. While modern tools address limitations in subjectivity and scalability, the core objective remains unchanged: using star counts to infer galactic structure.

5.3 Limitations

Several challenges persist:

- Bias toward brighter stars
- Difficulty resolving severe crowding
- Sensitivity to parameter tuning and data quality

5.4 Future Work

Potential improvements include:

- Enhanced preprocessing and noise modeling
- Improved blending models for crowded fields
- Completion and evaluation of CNN-based detection
- Calibration against external catalogs such as Gaia or Astrometry
- Application to multi-instrument and multi-wavelength datasets

Overall, this project demonstrates that modern computational tools can meaningfully extend Herschel’s original star gauging concept, enabling systematic and reproducible stellar density estimation with contemporary data.

5.5 Extra Information

All code and extra images can be viewed at my Github repository (updates are still currently in progress). https://github.com/CoolBro4811/ASTRAL_Project